

Case Study: Strategic Analysis of Q1 Travel Expense Reports

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1. Executive Summary

This analysis evaluates Q1 2024 travel expense claims with a focus on **process efficiency, compliance, and data quality**.

Key Findings

- **Average cycle time (submission → approval):** ~1.4 days
- **Late submissions (>6 months):** Present, including extreme outliers (>700 days)
- **Resubmission rate:** Low but non-zero
- **Data quality issues identified:**
 - Negative time differences
 - Extreme delays
 - Inconsistent timestamps

Key Insight

While average processing times appear efficient, **data inconsistencies and outliers significantly impact reliability**, highlighting the need for stronger validation and monitoring processes.

2. Analysis

A. Late Submission Analysis

Objective: Identify expense claims where the time between transaction date and submission date exceeds six months.

Methodology:

Delay calculated as:

- Submission Date – Transaction Date

Findings:

- Most claims are submitted within **1–30 days**
- However:
 - Multiple cases exceed **6 months**
 - Extreme outliers (>700 days) detected

Interpretation:

- Indicates occasional delays in reporting
 - Suggests potential issues in:
 - user behavior
 - system handling
 - data entry accuracy
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B. Cycle Time Analysis

Definition:

Cycle Time = First Submitted → First Approved

Findings:

- **Global average:** ~1.4 days
- Variation observed across countries

Key Observations:

- Majority of claims processed quickly
- Some regions show variability, indicating:
 - potential approval bottlenecks
 - organizational complexity

Important Note:

- Outliers significantly distort averages
 - **Median and percentile-based metrics are recommended** for more accurate reporting
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C. Resubmission Rate

Objective: Identify claims submitted multiple times.

Methodology:

Compare:

- First Submitted Date vs Last Submitted Date

Findings:

- Majority of claims submitted only once
- A small percentage required resubmission

Interpretation:

- Generally strong initial data entry quality
 - Resubmissions indicate:
 - validation gaps
 - policy misinterpretation
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D. Data Quality Assessment

Key Issues Identified:

- Negative submission delays (submission before transaction date)
- Extreme outliers (>700 days delay)
- Timestamp inconsistencies

Business Impact:

- Distorted KPIs
 - Misleading averages
 - Reduced confidence in reporting outputs
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3. Strategic Recommendations

1. Strengthen Data Validation Controls (High Priority)

- Enforce rules:
 - Submission date $\geq$ transaction date
 - Flag delays exceeding defined thresholds (e.g., 90 days)
 - Implement automated anomaly detection
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2. Optimize Approval Processes

- Analyze:
 - approval step count
 - approval time per step
 - Identify and address bottlenecks at role or region level
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3. Standardize KPI Definitions

- Clearly define:
 - cycle time calculation
 - inclusion/exclusion criteria
 - Use:
 - median
 - percentile metrics (P50, P90)
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4. Introduce Continuous Monitoring

- Develop dashboard including:
 - cycle time distribution
 - late submission trends
 - resubmission rates
 - data quality indicators
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5. Enable Data-Driven Cost Optimization (Optional)

- Perform spend concentration analysis
 - Identify high-volume vendors by region
 - Use insights to support vendor negotiations
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4. Conclusion

The analysis demonstrates that while operational efficiency is generally strong, **data quality issues and outliers introduce risk into performance measurement and decision-making.**

Addressing these gaps through **validation, monitoring, and standardized metrics** will significantly improve reporting accuracy and process transparency.

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